

Flowing Bits with the Lattice Boltzmann Method

From the BGK collision operator to the two-relaxation-time (TRT) method

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1 Introduction

Fluid dynamics looks deceptively simple. Water flows, air moves, smoke rises, and waves propagate. Yet describing these phenomena with mathematics is extremely difficult. The Navier–Stokes equations, which describe the motion of fluids such as water and air, have been known for more than a century, and their mathematical behavior in three dimensions still contains fundamental unanswered questions. In 2000, the Clay Mathematics Institute listed the Navier–Stokes existence and smoothness problem among its seven Millennium Prize Problems, with a prize of one million US dollars for a valid solution.

This is where computers enter the landscape. Modern machines can simulate fluids with remarkable detail and have been used in a wide range of applications, from weather prediction to aerodynamics, industrial flows, video games, and computer-generated imagery. A simulation, however, is not a solution. It is an approximation to a particular physical problem under particular assumptions, not a general mathematical solution of the equations. That is not a problem in practice: we do not need to solve the Navier–Stokes equations analytically before we can build useful simulations.

Computational fluid dynamics provides a family of numerical methods that approximate fluid behavior. Different methods make different compromises between physical accuracy, computational cost, numerical stability, and ease of implementation, such as classical grid-based methods, Smoothed Particle Hydrodynamics, and the Lattice Boltzmann Method (LBM).

LBM takes its own route. Instead of solving the macroscopic Navier–Stokes equations directly, it models the evolution of particle distribution functions on a discrete lattice. Local operations are repeated over many time steps, and the macroscopic fluid behavior emerges from these microscopic rules. The combination of local computation, regular memory access, and massive parallelism makes LBM a natural fit for modern computer architectures.

This report presents LBM from the ground up. It starts with the Bhatnagar–Gross–Krook (BGK) model and follows with the two-relaxation-time (TRT) method. It is written for anyone who had the fortune (or misfortune) of landing in a fluid dynamics or computational science course and who wants a clear idea of how a fluid emerges from a few local rules. Section 2 builds the mesoscopic picture and the BGK collision operator. Section 3 constructs the D2Q9 lattice. Section 4 states the full BGK scheme, derives its viscosity, and lists its limits. Section 5 introduces the TRT operator and its boundary correction. Appendix A collects scale tricks; Appendix B shows how a maze can be solved with a fluid simulation, an odd application I had for my solver.

2 From individual particles to a macroscopic view

2.1 Cloud of particles and continuum

A fluid is a cloud of particles. Each particle carries a position $\mathbf{x}$ and a velocity $\boldsymbol{\xi}$. A microscopic description tracks every particle. This is unusable for macroscopic flow: a milliliter of water holds on the order of 10^{21} molecules, and every collision changes the trajectories of all the particles involved. This would require N-Body simulations, which is far too expensive for practical large-scale fluid dynamics.

A continuum description goes to the opposite extreme. It replaces the cloud with a velocity field $\mathbf{u}(\mathbf{x})$ and a density field $\rho(\mathbf{x})$: one velocity and one density at every point of space. Two facts limit this picture. First, a continuum field is an infinite object: it assigns a value to every point of the domain, and a digital machine cannot store it or evolve it exactly. Every continuum simulation is a finite approximation of it. Second, the continuum equations are global. A Poisson equation determines the pressure of an incompressible flow; it couples every point to every other point, on every time step. The solver is expensive, and the mesh must conform to the solid walls.

LBM takes a middle path. It keeps the particles, but only in the form of their statistics. At each point of the lattice it stores how many particles move in each of a fixed set of directions. This is a mesoscopic description. A digital machine can represent it as it is a finite set of numbers per point. Its governing equation is local in space: the state of a point at the next step depends on the current state of that point and of its neighbors only. This locality is why LBM runs well on parallel machines. LBM is inspired by mesoscopic Boltzmann models, but the method does not solve the Boltzmann equation, and it does not need to [2].

2.2 The distribution function and its moments

We define the distribution function $f(\mathbf{x}, \boldsymbol{\xi}, t)$ as the number of particles at time t that are near position $\mathbf{x}$ and move with velocity near $\boldsymbol{\xi}$, per unit volume of phase space. It is a population count: f is non-negative, and the particles it counts are the ones arriving at $\mathbf{x}$ and the ones leaving $\mathbf{x}$. The state of a fluid cell is its mass, momentum, and stress, which is the content of f and of its velocity moments. Three moments carry the whole macroscopic state.

Density (zeroth moment).

$$\rho(\mathbf{x}, t) = \int f(\mathbf{x}, \boldsymbol{\xi}, t) d\boldsymbol{\xi}. \quad (1)$$

We sum f over all velocities and represent how much mass is at $\mathbf{x}$, the particles' directions do not matter.

Momentum (first moment).

$$\rho \mathbf{u}(\mathbf{x}, t) = \int f(\mathbf{x}, \boldsymbol{\xi}, t) \boldsymbol{\xi} d\boldsymbol{\xi}. \quad (2)$$

The integral weights each particle by its velocity and represents how much momentum is at $\mathbf{x}$, and in which direction. Division by ρ gives the mean velocity $\mathbf{u} = \rho^{-1} \int f \boldsymbol{\xi} d\boldsymbol{\xi}$. The momentum is a vector that points where the mass goes, and its magnitude says how much mass moves.

Stress (second moment).

$$\Pi_{\alpha\beta}(\mathbf{x}, t) = \int f(\mathbf{x}, \boldsymbol{\xi}, t) \xi_\alpha \xi_\beta d\boldsymbol{\xi}. \quad (3)$$

This is the momentum-flux tensor: component $\Pi_{\alpha\beta}$ is the flux of α -momentum across a surface whose normal is β . It measures how motion in one direction drags motion in the other. At local

equilibrium the flux splits into two parts,

$$\Pi_{\alpha\beta}^{\text{eq}} = p\delta_{\alpha\beta} + \rho u_\alpha u_\beta, \quad p = \rho c_s^2, \quad (4)$$

where p is the pressure and c_s the thermal speed scale (the speed of sound in this normalization). The term $p\delta_{\alpha\beta}$ is the isotropic pressure, the same flux in every direction. The term $\rho u_\alpha u_\beta$ is the convective flux which is the momentum that the bulk motion carries. Everything the fluid does is encoded in ρ , $\mathbf{u}$, and Π .

2.3 The Boltzmann equation

Two processes change f in time. A Free flight time where the particles move, and a Collisions time where the particles exchange momentum. The Boltzmann equation states their relation:

$$\frac{\partial f}{\partial t} + \boldsymbol{\xi} \cdot \nabla_{\mathbf{x}} f = \Omega(f), \quad \Omega(f) = \text{collision integral}. \quad (5)$$

The left-hand side is free flight, a particle with velocity $\boldsymbol{\xi}$ moves rigidly. The right-hand side, $\Omega(f)$, is the collision integral which sums the net rate at which pair scatterings bring particles into, and take them out of, the velocity $\boldsymbol{\xi}$. It is a high-dimensional, nonlinear integral over the velocity that is the main difficulty of fluid simulation.

The physics of collisions, however, is constrained. Collisions act fast and locally: They conserve mass and momentum but erase the shape of f and drive it toward the local equilibrium which is the Maxwell–Boltzmann distribution centered on the local state $(\rho, \mathbf{u})$,

$$f^{\text{eq}}(\boldsymbol{\xi}; \rho, \mathbf{u}) = \frac{\rho}{(2\pi c_s^2)^{D/2}} \exp\left(-\frac{|\boldsymbol{\xi} - \mathbf{u}|^2}{2 c_s^2}\right), \quad c_s^2 = \frac{k_B T}{m}, \quad (6)$$

where T is the absolute temperature, k_B the Boltzmann constant, and m the molecular mass. The quantity $c_s^2 = k_B T/m$ is the isothermal sound speed squared, and it equals the per-coordinate variance of the thermal velocity $|\boldsymbol{\xi} - \mathbf{u}|^2$; the Gaussian is normalized so that $\int f^{\text{eq}} d\boldsymbol{\xi} = \rho$. This is the same c_s^2 that sets the discrete sound speed of the lattice (for D2Q9, $c_s^2 = 1/3$). The equilibrium is the only distribution with no net collision effect, $\Omega(f^{\text{eq}}) = 0$. Collisions drag f toward f^{eq} ; the strength of the drag is set by the distance from f^{eq} .

The core of the *BGK approximation*, named after Bhatnagar, Gross and Krook [1], is to replace the full collision integral by one linear relaxation:

$$\Omega^{\text{BGK}}(f) = -\frac{1}{\tau}(f - f^{\text{eq}}). \quad (7)$$

In one unit of time, the fraction $1/\tau$ of the gap between f and f^{eq} is erased. The constant τ is the mean time between collisions, and it is the only parameter of the model. It is cheap to compute and also does not violate mass and momentum conservation.

$$\int \Omega^{\text{BGK}}(f) d\boldsymbol{\xi} = 0, \quad \int \boldsymbol{\xi} \Omega^{\text{BGK}}(f) d\boldsymbol{\xi} = 0. \quad (8)$$

Mass and momentum are conserved for every f .

2.4 Collide, then stream

We now discretize the velocity space: we keep a finite set of directions $\{\mathbf{c}_i\}_{i=0}^{Q-1}$, and one population f_i per direction. This in turns allows us to convert (5) to an explicit iterative algorithm. Every time step and at every lattice node:

Collide: relax each population toward its equilibrium, at the rate $1/\tau$.

Stream: move each population one cell along its own direction.

The collision step is local algebra: identical operations on independent cells. The streaming step is classical stencil operations. This pair of steps is the algorithmic core of every lattice Boltzmann scheme [2].

3 Introducing D2Q9

The collide–stream loop has two missing pieces: a set of lattice directions $\{\mathbf{c}_i\}$, and a discrete equilibrium f_i^{eq} to relax toward.

3.1 One-dimensional fluid

A one-dimensional gas lives on a line. A particle moves left, moves right, or sits still. The smallest useful velocity set has three members, at one lattice spacing c from the origin:

$$\mathbf{c}_i \in \{-c, 0, +c\}. \quad (9)$$

They represent moving left, right, or staying still.

At rest, the equilibrium is a constant, $f_i^{\text{eq}} = w_i \rho$, with weights w_i to be found. The weights must satisfy the moment identities of the one-dimensional Maxwellian (6). With c_s^2 the per-coordinate variance of that Maxwellian ($c_s^2 = p/\rho$):

$$\sum_i w_i = 1, \quad \sum_i w_i c_i = 0, \quad \sum_i w_i c_i^2 = c_s^2. \quad (10)$$

We can use the three-point Gauss–Hermite quadrature of a Gaussian to find the weights. Given a Gaussian and a target accuracy, it returns the points and the weights that reproduce the moments of the Gaussian up to a given order exactly. For a Gaussian of variance v it places the points at $\{0, \pm\sqrt{3}v\}$ with weights $\{2/3, 1/6, 1/6\}$. We set the variance to $c_s^2 = c^2/3$, so that the quadrature points are exactly $\{-c, 0, +c\}$. The weights are

$$w_- = w_+ = \frac{1}{6}, \quad w_0 = \frac{2}{3}. \quad (11)$$

We can check it: $\frac{1}{6} + \frac{2}{3} + \frac{1}{6} = 1$; $\frac{1}{6}(-c) + \frac{1}{6}(+c) = 0$; $\frac{1}{6}c^2 + \frac{1}{6}c^2 = \frac{1}{3}c^2 = c_s^2$. The ratio $c_s^2/c^2 = 1/3$ is the lattice sound speed; in lattice units $c = 1$ it reads $c_s^2 = 1/3$. At equilibrium, two thirds of the mass sits still, and one sixth of the mass flies in each direction. The weights are derived by the quadrature rule; they are not tuned parameters.

3.2 Moving up to 2D: the D2Q9 lattice

We now take the tensor product of the one-dimensional lattice with itself. Three velocities in x and three in y give $3 \times 3 = 9$ velocity states:

$$\mathbf{c}_i \in \{(0, 0), (1, 0), (0, 1), (-1, 0), (0, -1), (1, 1), (-1, 1), (-1, -1), (1, -1)\} \cdot c, \quad (12)$$

where c is the lattice speed (one lattice spacing per time step). The nine states split into three families: one rest state $(0, 0)$, four axis states at distance c , and four diagonal states at distance $\sqrt{2}c$. This configuration is called **D2Q9**: two dimensions, nine velocities. It is the classical two-dimensional lattice [2].

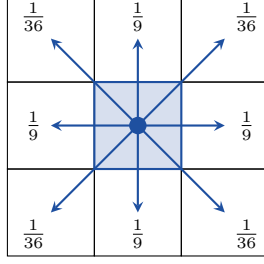


Figure 1: D2Q9 lattice: The center cell carries the rest population f_0 (weight $4/9$). The eight arrows show the streaming directions to the eight neighbors: four axis states at distance c (weight $1/9$) and four diagonal states at distance $\sqrt{2}c$ (weight $1/36$).

Evaluating the tensor product gives the canonical D2Q9 weights: $w_0 = 4/9$, $w_{1..4} = 1/9$, and $w_{5..8} = 1/36$, which satisfy the moment identities that make the lattice valid for a fluid.

$$\sum_i w_i = 1, \quad (13)$$

$$\sum_i w_i c_{i\alpha} = 0, \quad (14)$$

$$\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}, \quad c_s^2 = \frac{1}{3}, \quad (15)$$

$$\sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} = 0, \quad (16)$$

$$\sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} c_{i\delta} = c_s^4 (\delta_{\alpha\beta} \delta_{\gamma\delta} + \delta_{\alpha\gamma} \delta_{\beta\delta} + \delta_{\alpha\delta} \delta_{\beta\gamma}). \quad (17)$$

Identity (15) gives the lattice its speed of sound $c_s^2 = \frac{1}{3}$. Identity (17) is the one that matters: it is the isotropy¹ of the fourth velocity moment and state that the stress tensor the lattice can represent is rotationally invariant. A stencil with a single speed magnitude (the rest state plus the four axis states, for example) fails (17): it cannot supply the cross-coupling terms $\delta_{\alpha\gamma} \delta_{\beta\delta}$ and $\delta_{\alpha\delta} \delta_{\beta\gamma}$ ². The D2Q9 stencil is the smallest square stencil that passes.

3.3 Discrete equilibrium

The continuous equilibrium (6) has an infinite number of populations. The lattice keeps only the first few. We expand the Maxwellian in powers of the Mach number $\text{Ma} = |\mathbf{u}|/c_s \ll 1$ and truncate at order Ma^2 :

$$f_i^{\text{eq}}(\rho, \mathbf{u}) = w_i \rho \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2 - c_s^2 |\mathbf{u}|^2}{2 c_s^4} \right]. \quad (18)$$

The three terms correspond to different physical contributions. The term $w_i \rho$ is the rest population: it carries the density. The linear term in $\frac{w_i \rho}{c_s^2} (\mathbf{c}_i \cdot \mathbf{u})$ is the momentum: summing it over i with the identity (14) and (15) recovers $\rho \mathbf{u}$. The quadratic term is the convective flux $\rho u_\alpha u_\beta$: summing it with the fourth-moment identity (17) recovers the advective part of the stress (4). Dropping the quadratic term would leave the lattice unable to advect a moving fluid correctly; it is the term that makes the scheme a fluid scheme and not just a diffusion scheme³. Equation

¹Isotropy means that the properties of the lattice are the same in all directions.

²Because no particle flies diagonally, an axis-only stencil cannot physically carry momentum in the x -direction and y -direction at the same time within a single direction arrow. It lacks the cross-dimensional shear needed to make the fluid rotationally invariant.

³This term models the fact that the fluid has a bulk speed

(18) is the discrete equilibrium used by the BGK scheme and by the classical two-relaxation-time operator on D2Q9 [2, 4].

4 Bhatnagar-Gross-Krook scheme (BGK)

The lattice, discrete equilibrium, and collision rule now fixed, they together define the complete BGK scheme. In this section we will present its pseudocode and limitations.

4.1 The full scheme

In the split form, the BGK scheme is two local operations per node:

1. **Collision:**

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) + \frac{1}{\tau} (f_i^{\text{eq}}(\mathbf{x}, t) - f_i(\mathbf{x}, t))$$

2. **Streaming:**

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i^*(\mathbf{x}, t)$$

The collision step is the BGK operator (7) applied to the discrete populations. The streaming step is a permutation: population f_i at node $\mathbf{x}$ moves to the node $\mathbf{x} + \mathbf{c}_i$, where $\mathbf{c}_i$ is the direction. At solid walls the streaming step is replaced by a bounce-back: a population that would stream into a wall node is reflected, $f_{\bar{i}} \leftarrow f_i$, with $\mathbf{c}_{\bar{i}} = -\mathbf{c}_i$. Alternative on-site boundary conditions are explored in [3]. We examine that rule, and its weakness, in Sections 4.4 and 5.

The macroscopic state is read from the populations after each step:

$$\rho(\mathbf{x}, t) = \sum_{i=0}^8 f_i(\mathbf{x}, t), \quad \rho \mathbf{u}(\mathbf{x}, t) = \sum_{i=0}^8 f_i(\mathbf{x}, t) \mathbf{c}_i. \quad (19)$$

The density is the zeroth moment of the discrete distribution; the momentum is the first moment. The same moment-matching logic of Section 2.2 carries over verbatim to the discrete setting.

4.2 Pseudocode

```

1 initialize f[i] for all lattice nodes x, i = 0..8
2 repeat each time step:
3   for each node x:
4     rho = 0; ux = 0; uy = 0
5     for i = 0..8:
6       rho += f[i]
7       ux += f[i] * cx[i] // first moment
8       uy += f[i] * cy[i]
9     u = (ux/rho, uy/rho)
10    for i = 0..8: // equilibrium (eq. feq)
11      cu = cx[i]*ux/rho + cy[i]*uy/rho
12      u2 = ux*ux/rho/rho + uy*uy/rho/rho
13      feq[i] = w[i]*rho * (1 + cu/cs2 + (cu*cu - cs2*u2)/(2*cs2*cs2))
14    for i = 0..8: // collide
15      f[i] += tinv * (feq[i] - f[i]) // tinv = 1/tau
16    for i = 0..8: // stream
17      fnew[i] at x + c[i] = f[i]
18    apply bounce-back at solid walls
19    swap f and fnew
20 end simulation

```

Listing 1: BGK scheme on D2Q9, one time step.

4.3 Consistency verification

To verify that the loop solves a physical fluid, we use the Chapman–Enskog expansion. We order the populations in powers of the Knudsen number⁴: $f_i = f_i^{(0)} + Kn f_i^{(1)} + Kn^2 f_i^{(2)} + \dots$. And expand the streaming operator to second order. At order Kn^0 the populations sit at equilibrium, $f_i^{(0)} = f_i^{\text{eq}}$: the collision term vanishes. At order Kn^1 the moments give the Euler equations: mass conservation and momentum transport without viscosity. At order Kn^2 the slow time scale appears, and the deviatoric stress⁵ becomes proportional to the rate of strain⁶: the Newtonian viscous law. The scheme therefore reproduces the Navier–Stokes equations to second order, and its kinematic viscosity is fixed by the relaxation time:

$$\nu = c_s^2 \left(\tau - \frac{1}{2} \right) \Delta t. \quad (20)$$

For BGK, $s = 1/\tau$, so $\sigma = \tau - \frac{1}{2}$, where σ is the effective (or shifted) relaxation rate, usually called H  non’s parameter.

Three readings of (20) matter for what follows. First, τ is the viscosity control: a larger τ gives a larger ν , and the fluid is more damped. Second, the shift $-\frac{1}{2}$ is a discrete artifact. Streaming is a pure shift and carries no dissipation of its own; its second-order expansion contributes a term of $-\frac{1}{2}$ to the effective relaxation. The physical fluid is itself affected by the combination $\tau - \frac{1}{2}$, not τ alone. Third, ν must be positive, so

$$\tau > \frac{1}{2} \quad \Longleftrightarrow \quad s = \frac{1}{\tau} < 2. \quad (21)$$

The second form is the stability condition stated for lattice Boltzmann relaxation parameters [2]. Below $\tau = \frac{1}{2}$, the simulated viscosity is negative: the fluid amplifies gradients instead of damping them, and the solution grows without bound. This is one of the method’s limits.

One caveat, for completeness. The classical D2Q9 scheme with the equilibrium (18) approximates the incompressible Navier–Stokes equations up to second order in space and carries a third-order discrepancy in velocity; larger stencils (D2Q13, D2Q17) or modified equilibria remove it [2]. The discrepancy does not change any of the conclusions here as it is a higher-order correction.

4.4 The limits of one relaxation time

The BGK scheme has exactly one free parameter, τ . Every non-equilibrium mode of the nine-population distribution relaxes at that same rate. Three limits follow :

Limit 1: ceiling on the Reynolds number. A high-Reynolds flow wants a small viscosity. Equation (20) forces τ close to the stability threshold $\frac{1}{2}$ from above. The lattice carries more degrees of freedom than the three hydrodynamic fields: besides density and velocity there are non-hydrodynamic modes, higher velocity moments of the distribution, that carry no macroscopic information but are still evolved. As $\tau \rightarrow \frac{1}{2}^+$ their damping weakens, their amplitudes grow on the lattice time step, and the simulation diverges in a direction that the macroscopic fields do not show: the measured velocity and density look smooth, and then the grid explodes. In practice this caps the Reynolds number a single-rate scheme can reach. Multiple-relaxation-time schemes remove the cap by damping each mode at its own rate.

Limit 2: locked ratio of transport coefficients. In a thermal lattice Boltzmann model the momentum equation and the heat equation share the same relaxation rate. One rate sets both

⁴The Knudsen number indicates the validity of the continuum approximation.

⁵Internal friction caused by adjacent fluid layers sliding past each other. It pulls and twists the cube.

⁶Rate at which the cube is being distorted. If too high compared to simulation speed, it leads to local simulation instability, leading to breakdowns.

the momentum diffusivity and the thermal diffusivity, and their ratio, the Prandtl number⁷, is fixed to the lattice value. For BGK that value is $\text{Pr} = 1$. Real fluids do not adhere, for example : air has $\text{Pr} \approx 0.7$, water has $\text{Pr} \approx 7$. A single τ cannot set both diffusivities independently, so BGK cannot reproduce the transport behavior of a real fluid in a coupled thermal problem.

Limit 3: no-slip wall position. The bounce-back rule pins a no-slip point halfway between the last open node and the wall node, which is the geometric intent. In practice the effective wall position depends on τ . A pressure-driven channel flow computed at two viscosities gives two different effective widths; the measured discharge and permeability drift with the Reynolds number, and every no-slip boundary in the grid is a slip boundary displaced from its supposed location. This limit is the one the TRT operator is built to remove.

5 Two-relaxation-time (TRT) operator

The fix for the three limits above starts from one observation about the lattice: every direction has an opposite, the reversed direction. By exploiting that partner structure, and the single rate τ of BGK, we split the single rate into two independent rates.

The TRT method was first proposed by Ginzburg and d’Humières [5] in 2008.

5.1 Parity split

For every direction i , let $\bar{i}$ denote the reversed direction with $\mathbf{c}_{\bar{i}} = -\mathbf{c}_i$. The rest state $i = 0$ is its own partner ($\mathbf{c}_0 = \mathbf{0}$), while the remaining 8 directions form four moving pairs $i \in \{1, 2, 5, 6\}$ with opposites $\bar{i} \in \{3, 4, 7, 8\}$. Decomposing each population into even and odd parts:

$$f_i^+ = \frac{f_i + f_{\bar{i}}}{2}, \quad f_i^- = \frac{f_i - f_{\bar{i}}}{2}, \quad f_i = f_i^+ + f_i^-, \quad f_{\bar{i}} = f_i^+ - f_i^-. \quad (22)$$

Note that $f_0^+ = f_0$ and $f_0^- = 0$. Summing macroscopic quantities over direction pairs:

$$\rho = \sum_{i=0}^8 f_i = f_0 + \sum_{i \in \text{pairs}} (f_i + f_{\bar{i}}) = f_0^+ + 2 \sum_{i \in \text{pairs}} f_i^+, \quad (23)$$

$$\rho u_\alpha = \sum_{i=0}^8 f_i c_{i\alpha} = \sum_{i \in \text{pairs}} (f_i - f_{\bar{i}}) c_{i\alpha} = 2 \sum_{i \in \text{pairs}} f_i^- c_{i\alpha}, \quad (24)$$

$$\Pi_{\alpha\beta} = \sum_{i=0}^8 f_i c_{i\alpha} c_{i\beta} = \sum_{i \in \text{pairs}} (f_i + f_{\bar{i}}) c_{i\alpha} c_{i\beta} = 2 \sum_{i \in \text{pairs}} f_i^+ c_{i\alpha} c_{i\beta}, \quad (25)$$

where $\sum_{i \in \text{pairs}}$ sums over the four unique moving pairs. The density and stress live entirely in the even sector, while momentum lives entirely in the odd sector.

5.2 The TRT operator

Define the even and odd parts of the equilibrium, $f_i^{\text{eq}\pm} = (f_i^{\text{eq}} \pm f_{\bar{i}}^{\text{eq}})/2$, and relax each sector at its own rate:

$$\Omega_i^{\text{TRT}} = -\frac{1}{\tau_+} (f_i^+ - f_i^{\text{eq}+}) - \frac{1}{\tau_-} (f_i^- - f_i^{\text{eq}-}). \quad (26)$$

⁷The Prandtl number, $\text{Pr} = \nu/\alpha$, quantifies the ratio of momentum diffusivity (kinematic viscosity ν) to thermal diffusivity (α). In single-relaxation-time BGK models, a single relaxation time τ governs both momentum and energy transport, locking the model to a fixed $\text{Pr} = 1$. Simulating real fluids with non-unity Prandtl numbers (e.g., air at $\text{Pr} \approx 0.7$ or water at $\text{Pr} \approx 7$) requires multiple-relaxation-time schemes (such as TRT or MRT) to decouple viscous and thermal relaxation rates.

This is the two-relaxation-time operator which essentially is the BGK operator (7) with the single rate τ replaced by two rates acting on two orthogonal sectors.

Exact conservation, for any two rates. Collisions modify the total mass and momentum according to:

$$\begin{aligned}\sum_{i=0}^8 \Omega_i^{\text{TRT}} &= -\frac{1}{\tau^+}(f_0 - f_0^{\text{eq}}) - \frac{2}{\tau^+} \sum_{i \in \text{pairs}} (f_i^+ - f_i^{\text{eq}+}) \\ &= -\frac{1}{\tau^+} \left(\sum_{i=0}^8 f_i - \sum_{i=0}^8 f_i^{\text{eq}} \right) = -\frac{1}{\tau^+}(\rho - \rho) = 0,\end{aligned}\tag{27}$$

$$\sum_{i=0}^8 \mathbf{c}_i \Omega_i^{\text{TRT}} = -\frac{2}{\tau^-} \sum_{i \in \text{pairs}} (f_i^- - f_i^{\text{eq}-}) \mathbf{c}_i = -\frac{1}{\tau^-}(\rho \mathbf{u} - \rho \mathbf{u}) = \mathbf{0}.\tag{28}$$

Because the discrete equilibria preserve the zeroth and first moments exactly ($\sum_i f_i^{\text{eq}} = \rho$ and $\sum_i \mathbf{c}_i f_i^{\text{eq}} = \rho \mathbf{u}$), mass and momentum are unconditionally conserved for any choice of τ^+ and τ^- .

Two rates and one constraint. The stress lives in the even sector, so the viscosity is set by τ^+ :

$$\nu = c_s^2 \left(\tau^+ - \frac{1}{2} \right) \Delta t.\tag{29}$$

The odd rate τ^- is not part of the viscosity ν and only changes how the momentum sector and the non-hydrodynamic modes relax. In BGK the two rates were glued together, $\tau^+ = \tau^- = \tau$ whereas TRT separates them, giving it its name.

5.3 Boundary correction

If we redo the second-order boundary analysis of Section 4.4 for the half-way bounce-back rule, with two rates instead of one. The effective displacement of the no-slip point from its geometric location is proportional to one dimensionless combination of the two rates:

$$\delta_w \propto \left(\tau^+ - \frac{1}{2} \right) \left(\tau^- - \frac{1}{2} \right) - \frac{3}{16},\tag{30}$$

where the constant 3/16 is a property of the D2Q9 stencil combined with the half-way bounce-back rule. If we define the invariant

$$\Lambda = \left(\tau^+ - \frac{1}{2} \right) \left(\tau^- - \frac{1}{2} \right).\tag{31}$$

Under BGK, $\tau^- = \tau^+ = \tau$, so $\Lambda = (\tau - \frac{1}{2})^2$: the wall error (30) is generically non-zero, and it changes whenever the viscosity changes. This is Limit 3 of Section 4.4, in closed form.

Under TRT we choose

$$\tau^- = \frac{1}{2} + \frac{3/16}{\tau^+ - \frac{1}{2}} \iff \Lambda = \frac{3}{16},\tag{32}$$

and the wall error vanishes identically, $\delta_w \equiv 0$, for every value of τ^+ . One choice of τ^- pins the no-slip wall at its geometric location for all viscosities at once. The drift of Limit 3 is gone.

5.4 Pseudocode

The TRT simulation loop is very close to the BGK loop. The only change is the parity split and the two relaxation rates.

```

1 tau_plus = tau_p // sets the viscosity (eq. nu-trt)
2 tau_minus = 0.5 + (3.0/16.0)/(tau_p - 0.5) // pins the wall (eq. magic)
3 tplus = 1.0/tau_plus
4 tminus = 1.0/tau_minus
5 PAIRS = [(0, 0), (1, 3), (2, 4), (5, 7), (6, 8)]
6 // reversal table for D2Q9: 1<->3, 2<->4, 5<->7, 6<->8, 0<->0
7 repeat each time step:
8   for each node x:
9     rho, u, feq[i] as in BGK
10    for each pair (i, rev) in PAIRS:
11      fp = 0.5 * (f[i] + f[rev]) // even sector
12      fm = 0.5 * (f[i] - f[rev]) // odd sector
13      eqp = 0.5 * (feq[i] + feq[rev])
14      eqm = 0.5 * (feq[i] - feq[rev])
15      dp = tplus * (eqp - fp) // relax even sector
16      dm = tminus * (eqm - fm) // relax odd sector
17      f[i] = fp + fm + dp + dm
18      f[rev] = fp - fm + dp - dm
19    // stream and swap
20    for i = 0..8:
21      fnew[i] at (x + c[i]) = f[i]
22    apply bounce-back at solid walls
23    swap f and fnew
24 end simulation

```

Listing 2: TRT collision on D2Q9, one time step. The streaming step and the boundary conditions are unchanged from the BGK code.

5.5 What the TRT scheme does not fix

However, TRT is not perfect, and some limits remain.

Limit 1: the odd sector must stay damped. Both rates must lie in the stable range, $\tau^+ \geq \frac{1}{2}$ and $\tau^- \geq \frac{1}{2}$ (equivalently, each relaxation parameter stays below 2 [2]). The boundary correction (32) sends τ^- toward infinity as $\tau^+ \rightarrow \frac{1}{2}^+$, the frozen-odd-sector limit; the correction is then applied at the edge of the stable range, not in its interior. The knob works, but its useful range is bounded.

Limit 2: the correction is boundary-specific. The constant 3/16 in (30) belongs to the pair (D2Q9, half-way bounce-back, straight wall). The TRT wall is exact for the boundary rule it was derived for; it is not exact for every boundary rule.

Limit 3: one free rate is one free ratio. TRT gives one additional relaxation rate. It can tune one ratio of transport coefficients, but it cannot set all of them independently. In a coupled thermal model⁸ the Prandtl number is still constrained; reaching an arbitrary Pr needs a second population or a full multiple-relaxation-time operator. The bulk-to-shear viscosity ratio is also not freely settable. Limit 2 of BGK is eased but not closed.

Limit 4: the ghost modes share a rate. The odd sector damps all its non-hydrodynamic modes at the same rate τ^- , and the even sector does the same at τ^+ . A high-Reynolds flow wants the most unstable ghost mode damped hardest and the rest left alone; TRT cannot address modes individually. Multiple-relaxation-time schemes can, and at high Reynolds number they remain stable where two-rate schemes do not [4].

⁸Where heat can control flow, or flow can control heat.

A Scale tricks

A lattice simulation is dimensionless. The physical problem enters through a set of scale factors: a length scale Δx (meters per cell), a time scale Δt_p (seconds per lattice step), and the lattice speed c that they induce. A physical velocity U_p becomes the lattice velocity $U_l = U_p \Delta t_p / \Delta x$ (cells per step). A physical length L_p becomes the lattice length $L_l = L_p / \Delta x$ (cells). The Reynolds number is invariant under this rescaling:

$$\text{Re}_l = \frac{U_l L_l}{\nu_l} = \frac{U_p \Delta t_p / \Delta x \cdot L_p / \Delta x}{\nu_p \Delta t_p / \Delta x^2} = \frac{U_p L_p}{\nu_p} = \text{Re}_p. \quad (33)$$

The physics is unchanged when the scale changes. What does change is how the physics sits against the stability limits of the scheme, and those limits are stated in lattice units. Three bounds apply at once.

1. **Viscosity bound.** $\tau > \frac{1}{2}$, so that the simulated viscosity is positive (21). For D2Q9, $\tau = 3\nu_l + \frac{1}{2}$.
2. **Mach bound.** The compressibility error of the incompressible limit scales like $\text{Ma}^2 = (U_l/c_s)^2$, so one keeps $U_l/c_s \lesssim 0.1$.
3. **Courant-Friedrichs-Lewy (CFL) bound.** Streaming moves one cell per step, so $U_l \leq c = 1$.

The lattice velocity and the lattice viscosity are not independent. For a fixed physical problem they are tied by the invariant Reynolds number (33):

$$\nu_l = \frac{U_l L_l}{\text{Re}}. \quad (34)$$

The free choice is the target lattice velocity U_l (equivalently, the time step $\Delta t_p = U_l \Delta x / U_p$ for a given resolution). Choose U_l low enough that the Mach and CFL bounds hold; this always works, since U_l can be made arbitrarily small by taking Δt_p small enough. The cost is more time steps per unit physical time. The lattice viscosity is then fixed by the Reynolds number through (34). Lowering U_l therefore lowers ν_l in proportion, and the relaxation time $\tau = 3\nu_l + \frac{1}{2}$ is pushed toward the stability bound $\frac{1}{2}$. The two cannot be decoupled for a fixed physical problem: a low Mach number (needed for accuracy) forces a low viscosity (set by the physics), and both sit at the edge of the stable regime. For a high-Reynolds flow this tension is fundamental, because the Reynolds number is invariant under any rescaling.

Adaptive mesh refinement (AMR) sharpens the same idea. Refine the lattice only where the local velocity is high, such as a boundary layer, a jet, or a channel throat, and leave the rest coarse. AMR can become especially vital as the complexity of LBM increases in $\mathcal{O}(n^D)$, where D is the dimension of the simulation.

B Solving a maze with a fluid simulation

The following is a rather unconventional way to solve mazes, which are essentially connected networks of channels. A fluid pushed from one entrance toward another finds the path: the steady velocity field is non-zero only where a connected channel exists. The method uses that fact.

By pushing a flow through the entrance and waiting for an equilibrium to form, the pressures inside different channels of the maze will reach an equilibrium. Such equilibrium is dependent on the amount of fluid forced in maze. By looking at the velocity field once that state is reached, one can find a solution to the maze. An illustration is provided below.

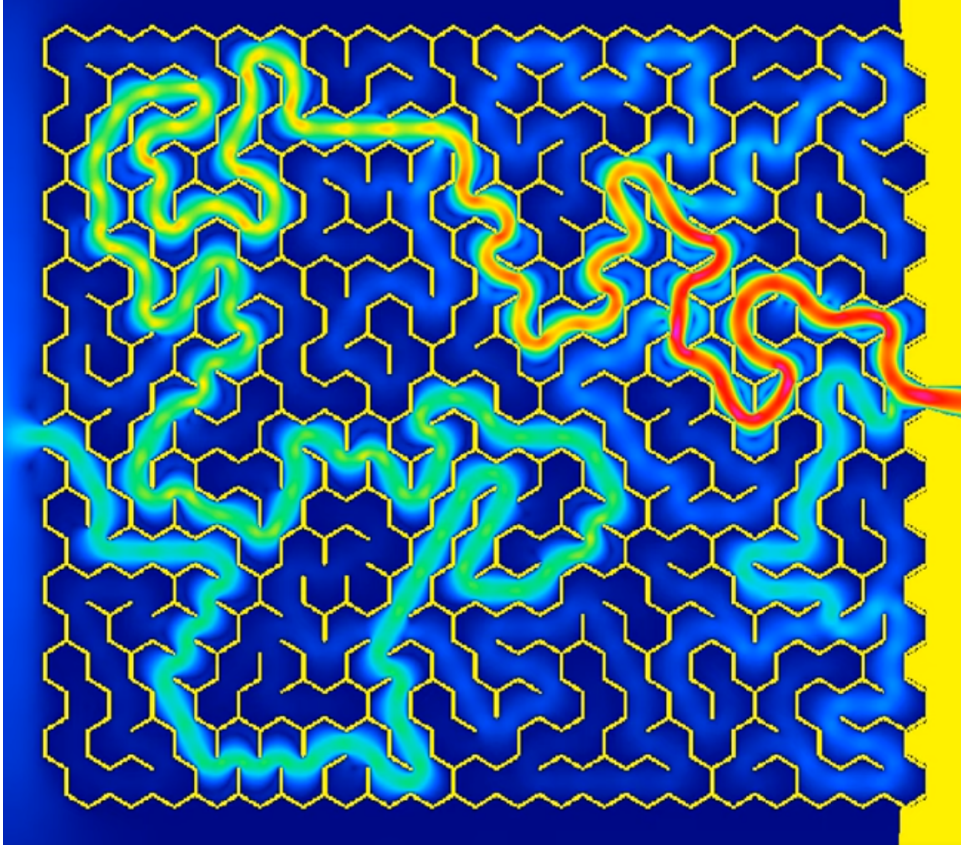


Figure 2: Velocity field after a quasi-equilibrium is reached inside a maze.

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